Fourth-Order Paired-Explicit Runge-Kutta Methods

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MS 147: Advances in Temporal Integration for High Order Methods

Daniel Doehring[1]

in collaboration with

L. Christmann[2], M. Schlottke-Lakemper[3],
G. Gassner[2], and M. Torrilhon[1]

[1] **ACoM, RWTH Aachen University** [2] CDS, University of Cologne [3] HPSC, University of Augsburg

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Applied and Computational Mathematics

RWTH AACHEN UNIVERSITY

Outline

Fourth-Order Paired-Explicit Runge-Kutta Methods

Scope

Consider PDEs of the form

$$\partial_t \mathbf{u}(t, \mathbf{x}) + \nabla \cdot \mathbf{f}\big(\mathbf{u}, \nabla \mathbf{u}\big) + \mathbf{g}(\mathbf{u}, \nabla \mathbf{u}) = \mathbf{s}\big(t, \mathbf{x}; \mathbf{u}\big)$$

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$$\frac{\mathrm{d}}{\mathrm{d}t}\mathbf{U}(t) = \mathbf{F}\big(t, \mathbf{U}(t)\big)$$

and target **convection-dominated** PDEs, i.e.,

$$\rho \sim \frac{|a|}{\Delta x} \gg \frac{d}{\Delta x^2}$$

where ρ is the spectral radius of

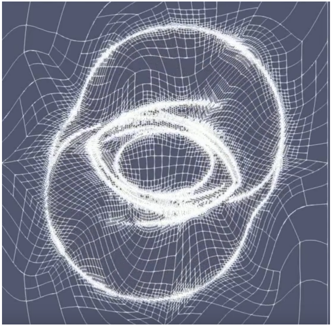
$$J_{\mathbf{F}}(\mathbf{U}) := \left.\frac{\partial \mathbf{F}}{\partial \mathbf{U}}\right|_{\mathbf{U}} .$$

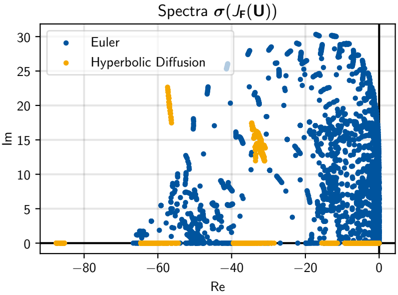
Motivation

Explicit time-integration for

1. non-uniform meshes
2. coupled systems

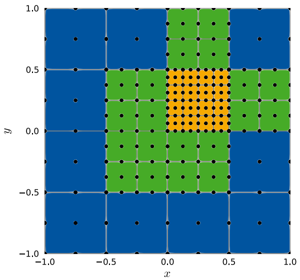
with **global** time-stepping leaves room for efficiency gains!

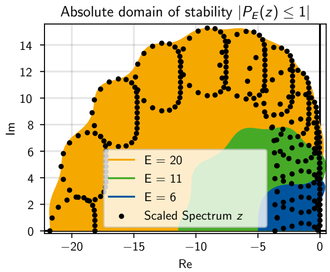




Partitioned Runge-Kutta Methods

Multirate time-integration by using **stabilized/optimized** schemes in regions with higher characteristic speeds (stricter CFL)





- For *convection-dominated* problems:

$$\Delta t \overset{!}{\leq} C_t(E) \cdot C_x(k) \min_{i,j} \frac{\Delta x_i}{|\mu_j(\mathbf{u}_i)|}, \quad \mu_j \in \sigma\left(\partial_{\mathbf{u}} f_j\right)$$

- For *optimized* Runge-Kutta schemes:

$$\Delta t_{p;E} \sim C_t(E) \sim E$$

Partitioned Runge-Kutta Methods

Partition ODE according to characteristic speeds $a_i := \min_j \frac{\Delta x_i}{|\mu_j(\mathbf{u}_i)|}$

$$\mathbf{U}'(t) = \begin{pmatrix} \mathbf{U}^{(1)}(t) \\ \vdots \\ \mathbf{U}^{(R)}(t) \end{pmatrix}' = \begin{pmatrix} \mathbf{F}^{(1)}\left(t, \mathbf{U}^{(1)}(t), \dots, \mathbf{U}^{(R)}(t)\right) \\ \vdots \\ \mathbf{F}^{(R)}\left(t, \mathbf{U}^{(1)}(t), \dots, \mathbf{U}^{(R)}(t)\right) \end{pmatrix} = \mathbf{F}\left(t, \mathbf{U}(t)\right)$$

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and employ different RKMs

$$\mathbf{K}_i^{(r)} = \mathbf{F}^{(r)} \left(t_n + c_i \Delta t, \mathbf{U}_n + \Delta t \sum_{j=1}^{S} \sum_{k=1}^{R} a_{i,j}^{(k)} \mathbf{K}_j^{(k)} \right),$$

$$\mathbf{U}_{n+1} = \mathbf{U}_n + \Delta t \sum_{i=1}^{S} b_i \sum_{k=1}^{R} \mathbf{K}_i^{(r)}$$

optimized for the spectrum $\boldsymbol{\sigma}\left(\frac{\partial \mathbf{F}}{\partial \mathbf{U}} \Big|_{\mathbf{U}_0} \right)$.

Partitioned Runge-Kutta Methods

- $S = 6,\ p = 2$ **Paired**[1] $E = \left\{E^{(1)}, E^{(2)}\right\} = \{3, 6\}$ Runge-Kutta

i	$\mathbf{c}$	$A^{(1)}$						$A^{(2)}$					
1	0												
2	$1/10$	$1/10$						$1/10$					
3	$2/10$	$2/10$	0					$2/10 - a^{(2)}_{3,2}$	$a^{(2)}_{3,2}$				
4	$3/10$	$3/10$	0	0				$3/10 - a^{(2)}_{4,3}$	0	$a^{(2)}_{4,3}$			
5	$4/10$	$4/10$	0	0	0			$4/10 - a^{(2)}_{5,4}$	0	0	$a^{(2)}_{5,4}$		
6	$5/10$	$5/10 - a^{(1)}_{6,5}$	0	0	0	$a^{(1)}_{6,5}$		$5/10 - a^{(2)}_{6,4}$	0	0	0	$a^{(2)}_{6,4}$	
	$\mathbf{b}^T$	0	0	0	0	0	1	0	0	0	0	0	1

[1] Vermeire; JCP; 2019

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6	$5/10$	$5/10 - a_{6,5}^{(1)}$	0	0	0	$a_{6,5}^{(1)}$		$5/10 - a_{6,4}^{(2)}$	0	0	0	$a_{6,4}^{(2)}$	
	$\mathbf{b}^T$	0	0	0	0	0	1	0	0	0	0	0	1

- Method $r = 1$ requires $\mathbf{F}^{(1)}$ *evaluation only for* $i = \mathbf{1}, \mathbf{5}, \mathbf{6}$

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– Conservation: $\mathbf{b}^{(r)} \equiv \mathbf{b}\ \forall r$

– Consistency: $\mathbf{c}^{(r)} \equiv \mathbf{c}\ \forall r$

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Optimized Stability Polynomials

Objective: Maximize *linear* stability

$$\max_{P_{p;E} \in \mathcal{P}_{p;E}} \Delta t \text{ such that } \left|P_{p;E}(\Delta t \lambda_m)\right| \leq 1, \quad \lambda_m \in \boldsymbol{\sigma}\left(J_{\mathbf{F}}(\mathbf{U}_0)\right)$$

[1] Ketcheson & Ahmadia; CAMCoS; 2012
[2] D., Gassner, Torrilhon; J. Sci. Comput.; 2024

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where $\mathcal{P}_{p;E}$ denotes the set of **stability polynomials** of (linear) order p and degree E, i.e.,

$$P_{p;E}(z;\alpha) = \sum_{j=0}^{p} \frac{1}{j!} z^j + \sum_{j=p+1}^{p+E} \alpha_j z^j$$

which allows for efficient optimization.[1]

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which allows for efficient optimization.[1]

– Alternatively (robust for high E)[2]

$$P_{p;E}(z;\widetilde{\mathbf{r}}) = 1 + z \prod_{j=1}^{E-1} \left(1 - \frac{z}{\widetilde{r}_j}\right)$$

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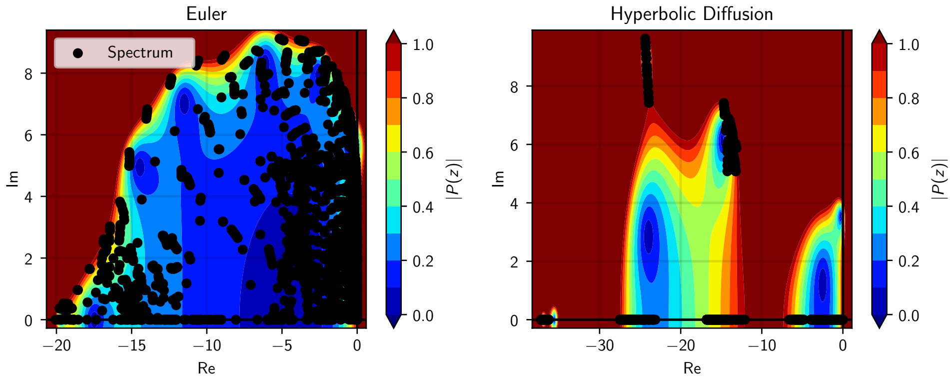

Optimized Schemes: Constraints

- **Construct** Runge-Kutta method (Butcher tableau $A^{(r)}, \mathbf{b}^T, \mathbf{c}$) from stability polynomial $P_{p;E}(z)$ *such that*

$$P^{(r)}_{p;E}(z) \overset{!}{=} 1 + z\mathbf{b}^T \left(I - zA^{(r)}\right)^{-1} \mathbf{1} \quad \text{(Linear Stability)}$$

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Order Conditions

$p = 3$

$$\mathbf{b}^T \mathbf{1} \overset{!}{=} 1$$

$$\mathbf{b}^T \mathbf{c} \overset{!}{=} \frac{1}{2}$$

$$\mathbf{b}^T \mathbf{c}^2 \overset{!}{=} \frac{1}{3}$$

$$\mathbf{b}^T A^{(r)} \mathbf{c} \overset{!}{=} \frac{1}{6} \quad \forall\, r$$

$p = 4$

$$\mathbf{b}^T \mathbf{c}^3 \overset{!}{=} \frac{1}{4}$$

$$\mathbf{b}^T C A^{(r)} \mathbf{c} \overset{!}{=} \frac{1}{8} \quad \forall\, r$$

$$\mathbf{b}^T A^{(r)} \mathbf{c}^2 \overset{!}{=} \frac{1}{12} \quad \forall\, r$$

$$\mathbf{b}^T A^{(r_1)} A^{(r_2)} \mathbf{c} \overset{!}{=} \frac{1}{24} \quad \forall\, r_1, r_2$$

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$$\mathbf{b}^T A^{(r_1)} A^{(r_2)} \mathbf{c} \overset{!}{=} \frac{1}{24} \quad \forall\, r_1, r_2$$

and form of A (reducible property) +
conservation & internal consistency (shared $\mathbf{b}^T, \mathbf{c}$) are kept!

Constructing 4th-Order Schemes

Central Ideas:

- Maximally sparse $\mathbf{b}^T$: Only $b_{S-1}, b_S \neq 0$
  $\Rightarrow$ Maintain reducibility, simplify optimization & construction
- Shared last 4 rows of $A^{(r)}$ to fulfill $\mathbf{b}^T A^{(r_1)} A^{(r_2)} \mathbf{c} \quad \forall\, r_1, r_2$
  $\Rightarrow$ Use remaining rows to match stability polynomial

Keep "P-ERK layout" of A:

$$a_{i,i-1} \neq 0, \quad a_{i,1} = c_i - a_{i,i-1}$$

Outcome:

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Outcome:

$\Rightarrow$ 9 Unknowns, 8 Equations for last 4 rows
$\Rightarrow$ 1 – Parameter (c_{S-3}) solution!
$\Rightarrow$ Can compute remaining $a_{i,i-1}$ directly from $P_{p;E}(z)$

Constructing 4th-Order Schemes

$$
\begin{array}{c|c|ccc}
 & & & r & \\ \hline
 & & 1 & \dots & R \\ \cline{3-5}
i & \mathbf{c} & A^{(1)} & \dots & A^{(R)} \\ \hline
 & & & \mathbf{b}^T &
\end{array}
=
\begin{array}{c|c|ccc}
 & & & r & \\ \hline
i & \mathbf{c} & 1 & \dots & R \\ \hline
1 & 0 & & & \\
2 & c_2 & & A_{1:2} & \\ \hdashline
3 & c_3 & & & \\
\vdots & \vdots & A^{(1)}_{3:S-4} & \dots & A^{(R)}_{3:S-4} \\
S-4 & c_{S-4} & & & \\ \hdashline
S-3 & c_{S-3} & & & \\
\vdots & \vdots & & A_{S-3:S} & \\
S & c_S & & & \\ \hline
 & & & \mathbf{b}^T &
\end{array}
$$

P-ERK $\mathbf{p} = \mathbf{4}$: Loss of Optimality

Pay the price for unifiying last 4 rows $A_{S-3:S}$:

- Require **5** stages for an explicit method
- Fixed α_5 coefficient of $P_{4;E}(z)$

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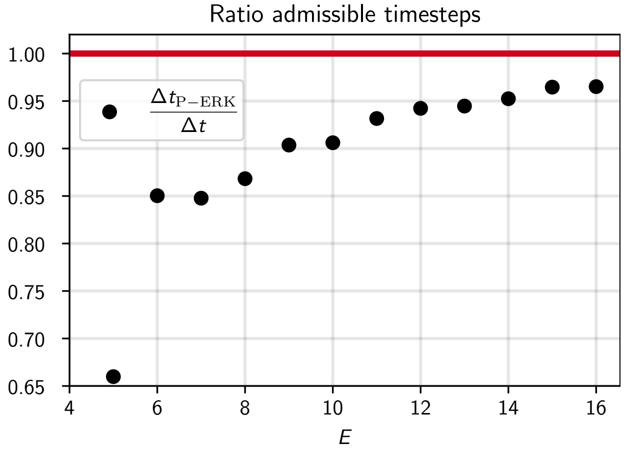
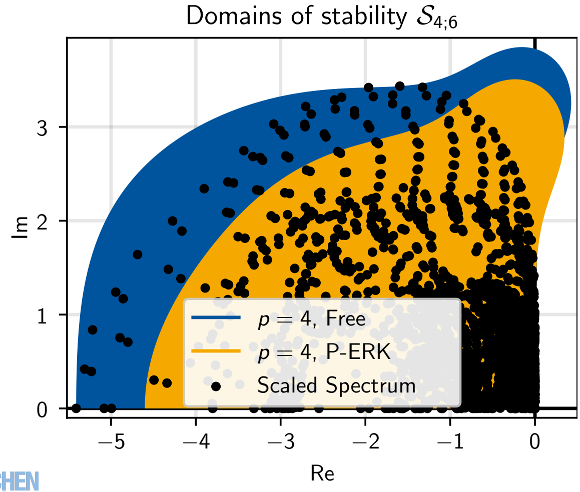


P-ERK $p = 4$: Loss of Optimality

Pay the price for being lazy before:

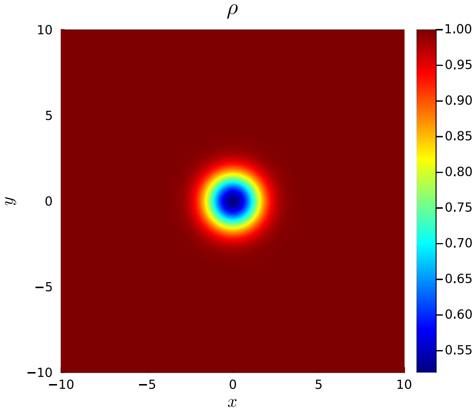
- Require **5** stages for an explicit method
- Fixed α_5 coefficient of $P_{4;E}(z)$

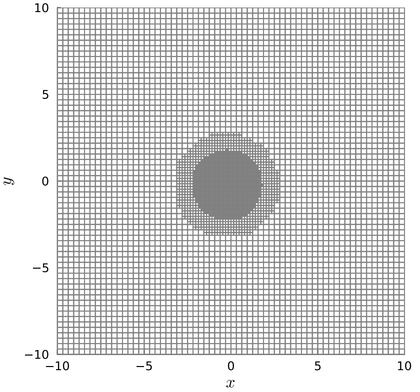


Validation: Convergence

Isentropic Vortex

- DGSEM, $k = 6$, HLLC, weak-form volume integral
- 64^2 base discretization, 3-Level AMR





Validation: Convergence

Isentropic Vortex

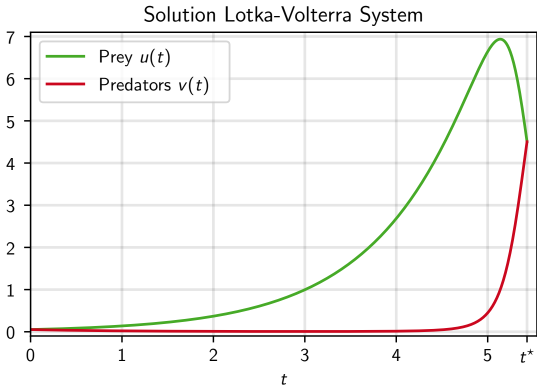
- $k = 6$, HLLC, weak-form volume integral
- 64^2 base discretization, 3-Level AMR
- $E = \{5, 9, 15\}$ P-ERK scheme

CFL	$\frac{1}{\vert\Omega\vert}\Vert e_\rho(x,y)\Vert_{L^1(\Omega)}$	EOC	$\Vert e_\rho(x,y)\Vert_{L^\infty(\Omega)}$	EOC
1	$3.37848279 \cdot 10^{-10}$		$1.30746968 \cdot 10^{-6}$	
$1/2$	$2.12288265 \cdot 10^{-11}$	3.99	$4.23331622 \cdot 10^{-8}$	4.95
$1/4$	$6.47315254 \cdot 10^{-12}$	1.71	$1.96120620 \cdot 10^{-9}$	4.43

Validation: Convergence

Lotka-Volterra

- $\dot{u}(t) = u(1 - v), \; \dot{v}(t) = v(u - 1) \quad \Rightarrow R = 2$
- Solution involving quadrature[1] for $t \in [0, t^\star)$



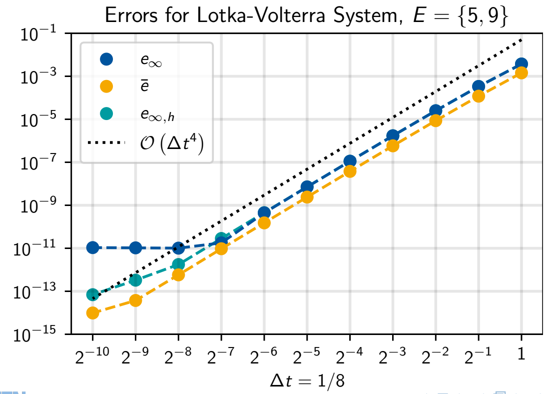
[1] Boulnois; ArXiv; 2023

Validation: Convergence

Lotka-Volterra

- $\dot{u}(t) = u(1-v), \ \dot{v}(t) = v(u-1)$
- Solution involving quadrature

$$e_\infty := \left\| \begin{pmatrix} u_h(t^\star) \\ v_h(t^\star) \end{pmatrix} - \begin{pmatrix} u(t^\star) \\ v(t^\star) \end{pmatrix} \right\|_\infty \qquad \bar{e} := \left| \frac{u_h(t^\star) + v_h(t^\star)}{2} - \frac{u(t^\star) + v(t^\star)}{2} \right|$$



Validation: Conservation

Isentropic Vortex

- DGSEM, $k = 3$, HLLC, weak-form volume integral
- 64^2 base discretization, 3-Level AMR

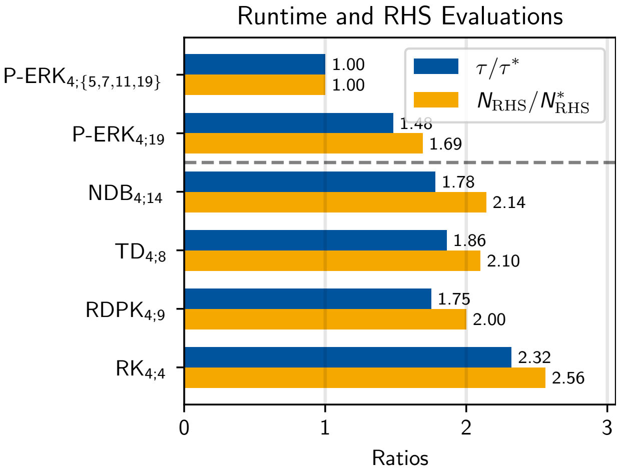
$$e_u^{\text{Cons}}(t_f) := \left| \int_\Omega u(t_0, \mathbf{x}) \mathrm{d}\mathbf{x} - \int_\Omega u(t_f, \mathbf{x}) \mathrm{d}\mathbf{x} \right|$$

	P-ERK$_{4;\{6,10,16\}}$	SSP$_{4;10}$
$e_\rho^{\text{Cons}}(t_f)$	$3.78 \cdot 10^{-13}$	$2.11 \cdot 10^{-13}$
$e_{\rho v_x}^{\text{Cons}}(t_f)$	$5.70 \cdot 10^{-14}$	$1.26 \cdot 10^{-13}$
$e_{\rho v_y}^{\text{Cons}}(t_f)$	$6.53 \cdot 10^{-14}$	$1.29 \cdot 10^{-13}$
$e_{\rho e}^{\text{Cons}}(t_f)$	$9.40 \cdot 10^{-13}$	$3.66 \cdot 10^{-12}$

Application: Non-Uniform Meshes

Isentropic Vortex

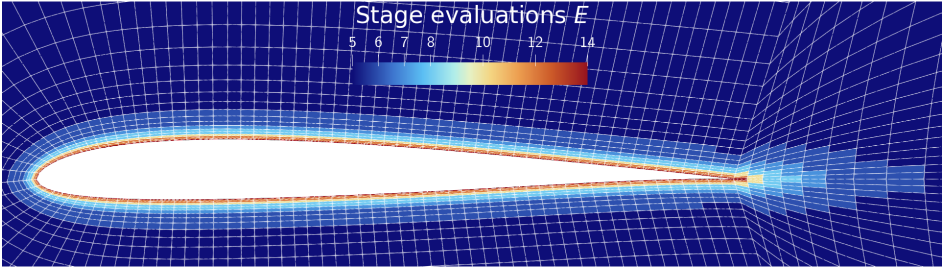
- DGSEM, $k = 3$, HLLC, weak-form volume integral
- 64^2 base discretization, 4-Level AMR



Application: Non-Uniform Meshes

SD7003 Airfoil

- Ma $= 0.2$, Re $= 10^4$, Pr $= 0.72$, $\alpha = 4°$
- $k = 3$, HLLC, Flux-Differencing with Ranocha-Flux, BR1
- $\Omega = [-20, 40] \times [-20, 20]$, 7605 linear quads



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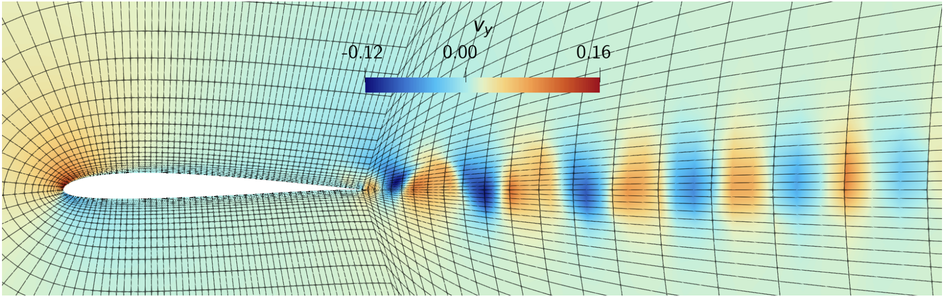

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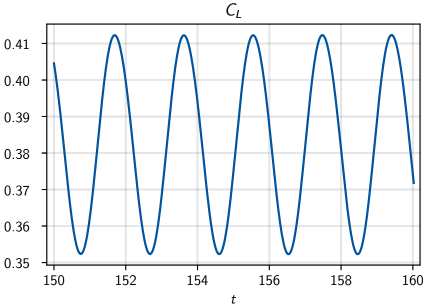
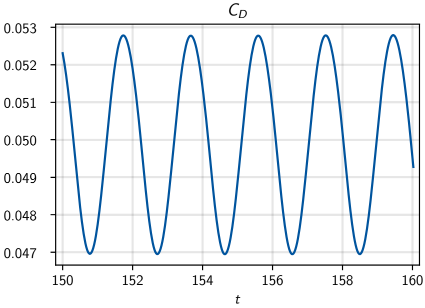


Lift coefficient $C_L = C_{L,p}$



Drag coefficient $C_D = C_{D,p} + C_{D,\mu}$

Application: Non-Uniform Meshes

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Source	$\overline{C}_L$	$\overline{C}_D$
P-ERK $p = 4$	0.3827	0.04995
P-ERK $p = 2$[1]	0.3841	0.04990
P-ERK $p = 3$[2]	0.3848	0.04910
Uranga et al.[3]	0.3755	0.04978
López-Morales et al.[4]	0.3719	0.04940

[1] Vermeire. JCP. 2019

[2] Nasab, Vermeire. JCP. 2022

[3] Uranga et al. Int. J. Num Methods Eng. 2011

[4] López-Morales et al. 32nd AIAA AAC. 2014

Application: Non-Uniform Meshes

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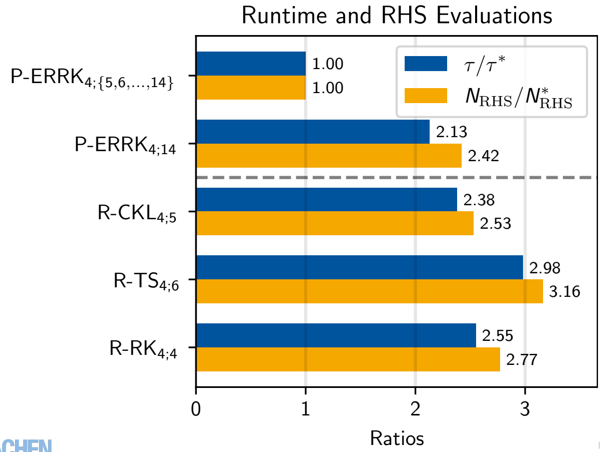
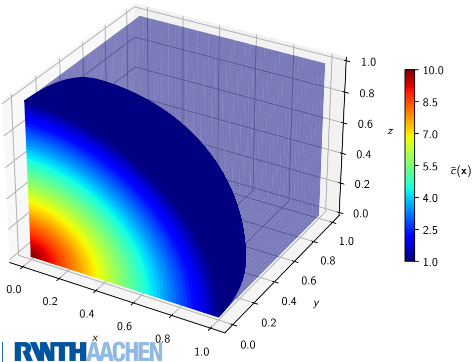


Application: Non-Uniform Characteristic Speeds

3D Linearized Euler with non-uniform mean speed of sound $\bar{c}(\mathbf{x})$

$$\partial_t \begin{pmatrix} \rho' \\ v_1' \\ v_2' \\ v_3' \\ p' \end{pmatrix} + \partial_x \begin{pmatrix} \bar{\rho} v_1' + \overline{v_1} \rho' \\ \overline{v_1} v_1' + \frac{p'}{\bar{\rho}} \\ \overline{v_1} v_2' \\ \overline{v_1} v_3' \\ \overline{v_1} p' + \bar{c}^2 \bar{\rho} v_1' \end{pmatrix} + \partial_y \begin{pmatrix} \bar{\rho} v_2' + \overline{v_2} \rho' \\ \overline{v_2} v_1' \\ \overline{v_2} v_2' + \frac{p'}{\bar{\rho}} \\ \overline{v_2} v_3' \\ \overline{v_2} p' + \bar{c}^2 \bar{\rho} v_2' \end{pmatrix} + \partial_z \begin{pmatrix} \bar{\rho} v_3' + \overline{v_3} \rho' \\ \overline{v_3} v_1' \\ \overline{v_3} v_2' \\ \overline{v_3} v_3' + \frac{p'}{\bar{\rho}} \\ \overline{v_3} p' + \bar{c}^2 \bar{\rho} v_3' \end{pmatrix} = \mathbf{0}$$



Characteristic Speeds:

$$\mu_j(\mathbf{x}) = \bar{v}_j + \bar{c}(\mathbf{x})$$

Initialization: Acoustic wave:

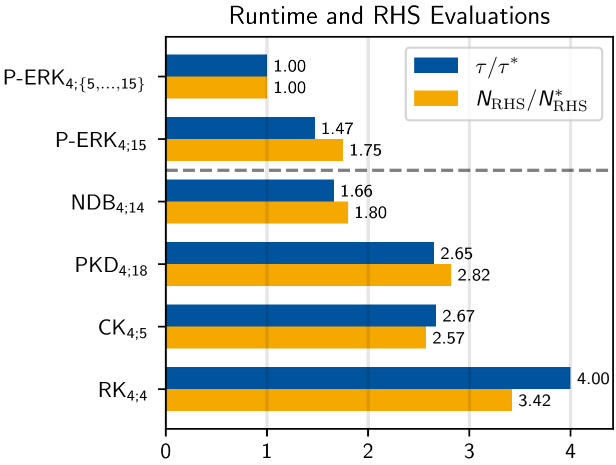
$v_j'(0) = \exp\left(-30\|\mathbf{x}\|_2^2\right)$,

$\rho'(0) = p'(0) = -v_j'(0)$.

Application: Non-Uniform Characteristic Speeds

3D Linearized Euler with non-uniform mean speed of sound $\bar{c}(\mathbf{x})$

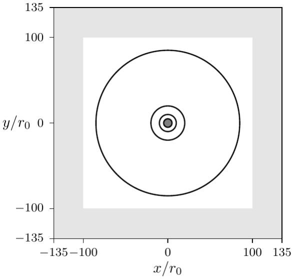
- 84 Million DoF

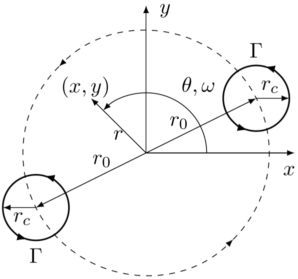


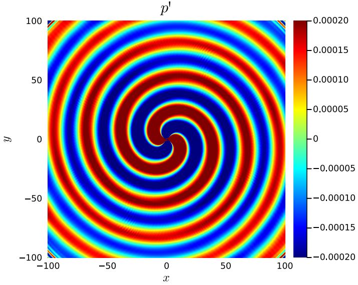
Application: (Bulk) Coupled Systems

Euler-Acoustic: Use (Nonlinear) Euler/CFD solver to compute source terms for the (linear) acoustic perturbation solver

- Sound generated by rotating vortices[1]







[1] Schlottke-Lakemper et al.; Comput. Fluids; 2017

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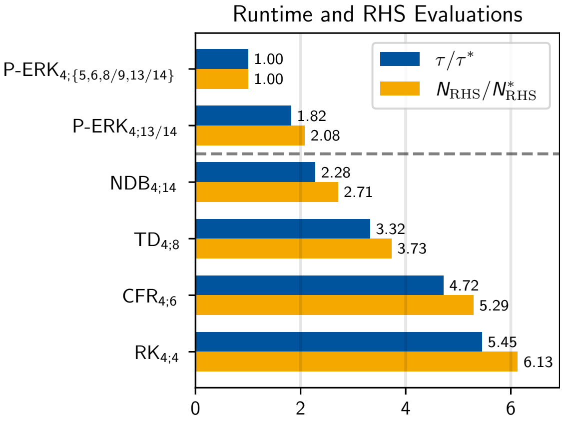

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Conclusion & Outlook

Construction of 4^{th}-Order P-ERK Schemes

- Sparse $\mathbf{b}^T$ & shared part of $A^{(r)}$ to satisfy *coupled nonlinear* order conditions
- $S_{\text{min}} = 5 \Rightarrow$ slight loss of optimality

Conclusion & Outlook

Construction of 4^{th}-Order P-ERK Schemes

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- $S_{\min} = 5 \Rightarrow$ slight loss of optimality

Applications & Related Work:

- Non-Uniform meshes/char. speeds, potentially with AMR
- Bulk-Coupled Systems: Euler-Acoustic, Euler-Gravity
- Entropy-Conservation via Relaxation

  $\Rightarrow$ Talk tomorrow part of MS 132

Thank You for your attention!

Questions?

doehring@acom.rwth-aachen.de
https://www.acom.rwth-aachen.de
https://github.com/DanielDoehring
https://arxiv.org/abs/2408.05470

Reproducibility Repository: https://github.com/DanielDoehring/paper-2024-perk4

Trixi.jl: https://github.com/trixi-framework/Trixi.jl

Application: Euler-Gravity

Euler (2D)

$$\partial_t \begin{pmatrix} \rho \\ \rho v_x \\ \rho v_y \\ E \end{pmatrix} + \nabla \cdot \begin{pmatrix} \rho v_x & \rho v_y \\ \rho v_x^2 + p & \rho v_x v_y \\ \rho v_x v_y & \rho v_y^2 + p \\ (E+p) v_x & (E+p) v_y \end{pmatrix}$$

[1] Schlottke-Lakemper et al.; JCP; 2021

Application: Euler-Gravity

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+ a **self-gravitational potential** ϕ

$$-\Delta \phi = -4\pi G \rho$$

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\+ a **self-gravitational potential** ϕ

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which *couples* to the momentum and energy equations[1].

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Application: Euler-Gravity

Euler (2D) (hyperbolic)

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$+$ a **self-gravitational potential** ϕ (elliptic)

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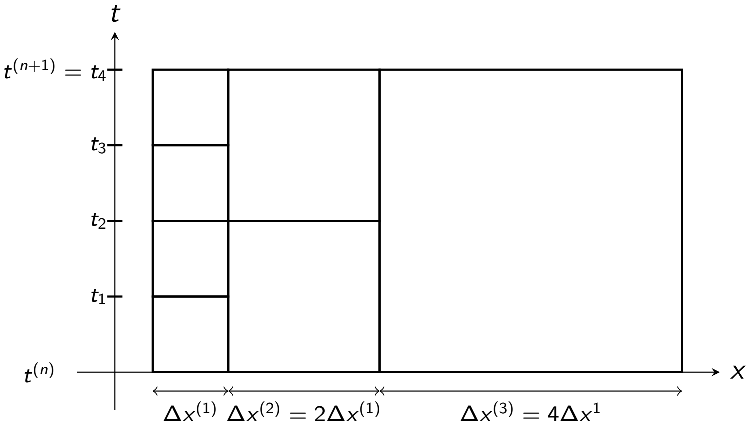
Application: Euler-Gravity

- **Hyperbolic – Elliptic coupling** for *Local Time-Stepping*



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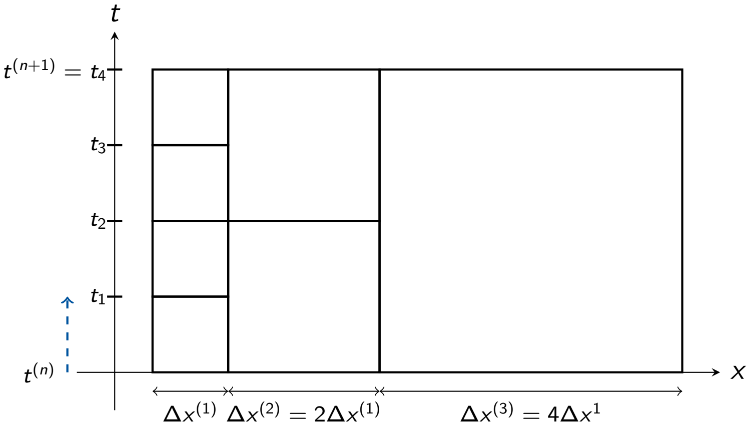

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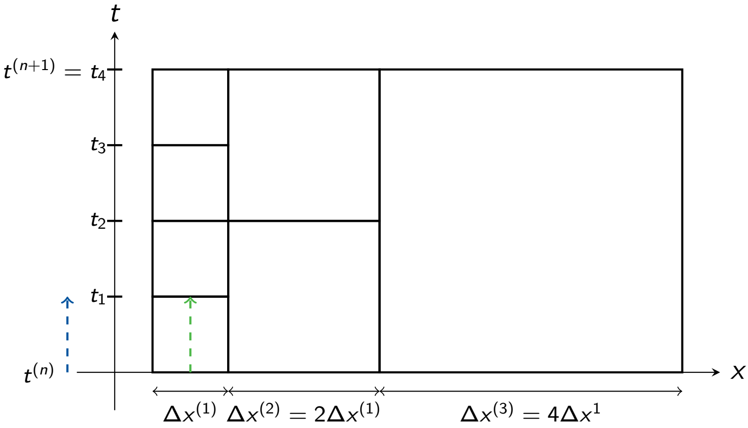

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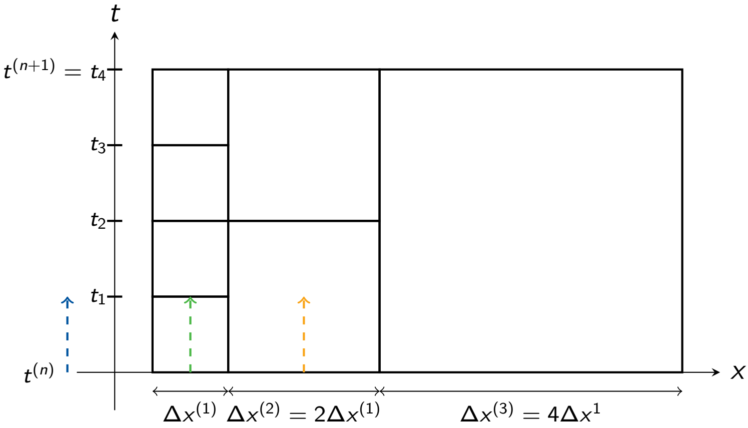

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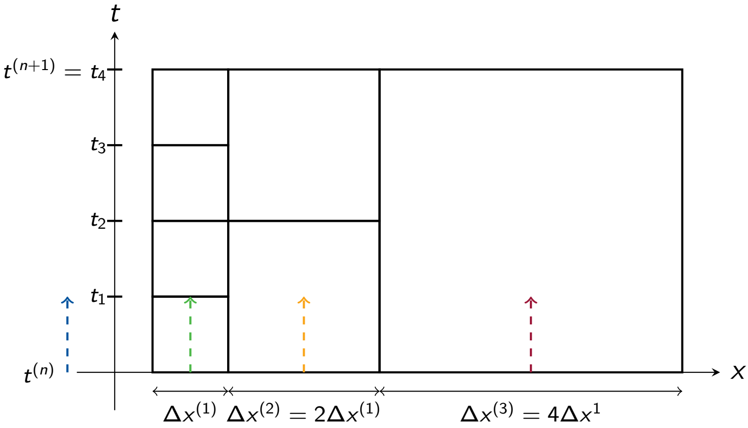

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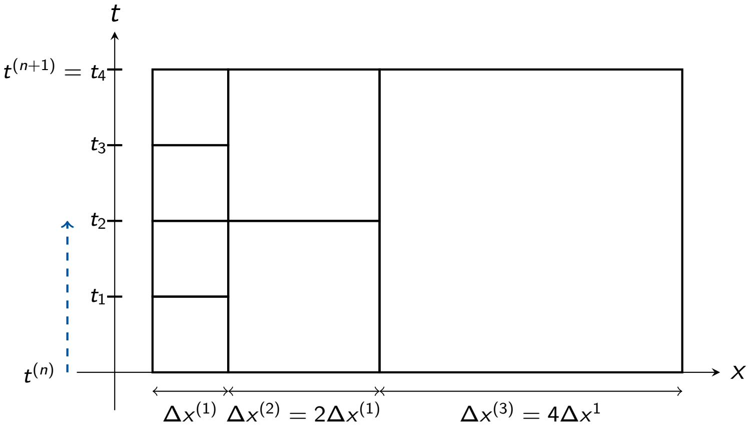

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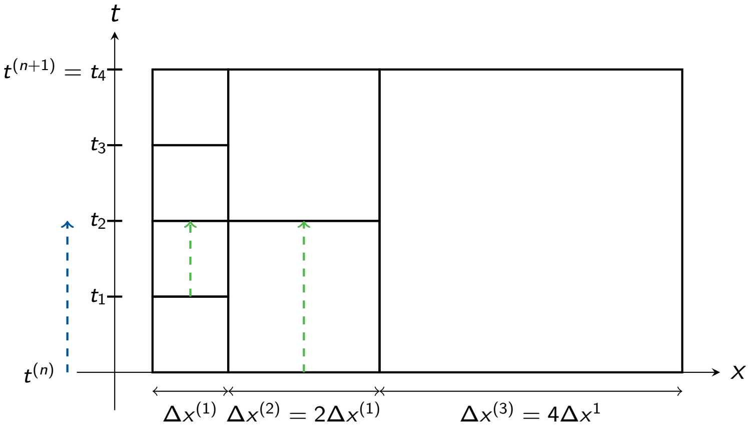

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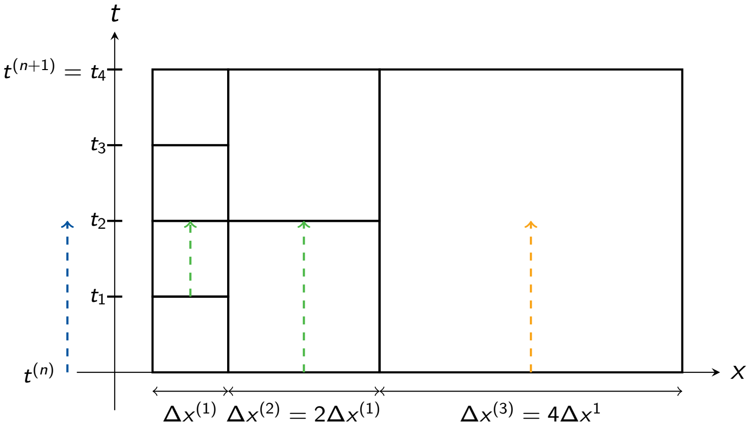

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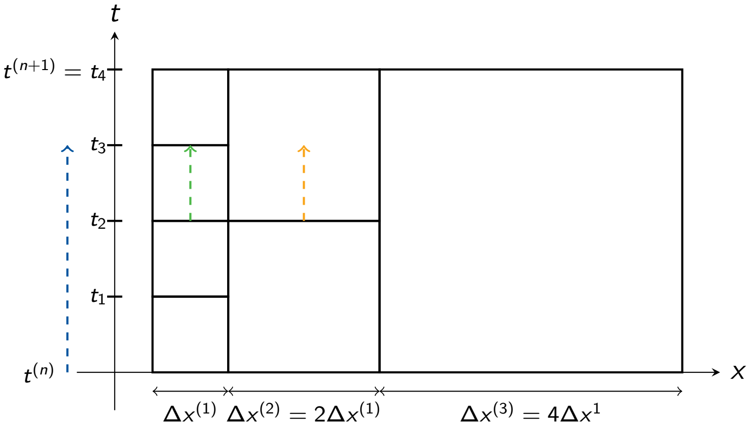

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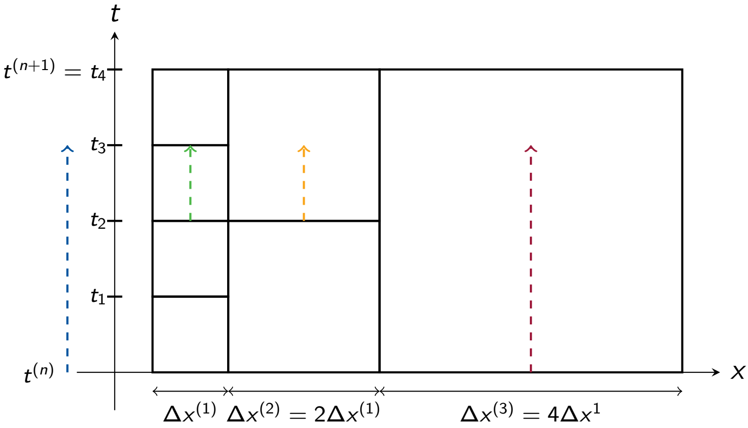

Application: Euler-Gravity

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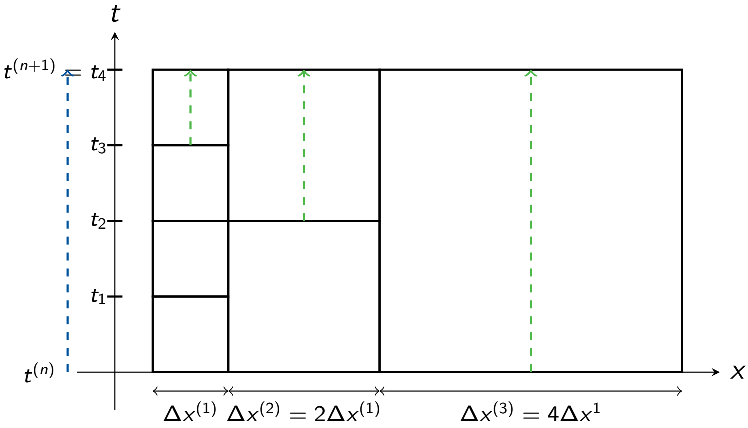

Application: Euler-Gravity

Need to solve

$$-\Delta \phi = -4\pi G \rho$$

after every update to ρ.

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$$\rightsquigarrow \underbrace{\partial_\tau \begin{pmatrix} \phi \\ q_1 \\ q_2 \end{pmatrix} + \nabla \cdot \begin{pmatrix} -q_1 & -q_2 \\ -\phi/T_r & 0 \\ 0 & -\phi/T_r \end{pmatrix} = \begin{pmatrix} -4\pi G\rho \\ -q_1/T_r \\ -q_2/T_r \end{pmatrix}}_{\textit{Hyperbolic Diffusion Equations}}, \qquad \begin{aligned} \mathbf{q} &= \nabla\phi \\ T_r &= \text{const} \end{aligned}$$

after every update to ρ.

Application: Euler-Gravity

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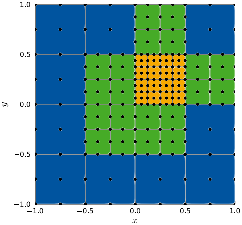

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$$\mathbf{q} = \nabla\phi$$

$$T_r = \text{const}$$

after every update to ρ.

- P-ERK schemes: Always at <u>same time-level</u> $t = t_n + c_i\Delta t$!

$$\mathbf{c} = \begin{array}{c|c|c|c} c_1 & & & \\ \vdots & A^{(1)} & A^{(2)} & A^{(3)} \\ c_S & & & \\ \hline & & \mathbf{b}^T & \end{array}$$

Application: Euler-Gravity

- **Convergence Study**

$$\rho = 2 + \frac{1}{10} \sin\left(\pi(x+y-t)\right)$$

$$v_x = v_y = 1$$

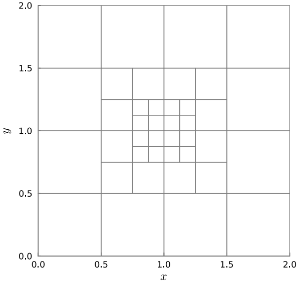
$$p = \frac{\rho^2}{\pi^2}$$

$$\phi = \frac{2}{\pi}(2-\rho)$$

$+$ source terms (manufactured solution)

3-Level Grid

$\Rightarrow$ 3-Level methods for Euler & Hyp.–Diff.



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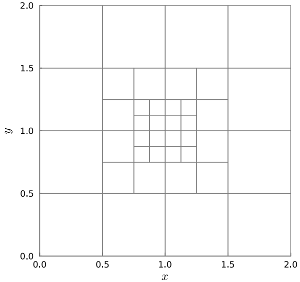

N	$L^\infty(\rho)$	$L^\infty(v_{x,y})$	$L^\infty(p)$	$L^\infty(\phi)$	$L^\infty(q_{1,2})$
40	$2.5 \cdot 10^{-4}$	$5.2 \cdot 10^{-4}$	$3.0 \cdot 10^{-3}$	$1.5 \cdot 10^{-3}$	$3.8 \cdot 10^{-3}$
160	$1.4 \cdot 10^{-4}$	$4.4 \cdot 10^{-5}$	$1.6 \cdot 10^{-4}$	$7.4 \cdot 10^{-5}$	$2.8 \cdot 10^{-4}$
640	$6.6 \cdot 10^{-6}$	$3.0 \cdot 10^{-6}$	$1.0 \cdot 10^{-5}$	$7.8 \cdot 10^{-6}$	$3.2 \cdot 10^{-5}$
2560	$4.2 \cdot 10^{-7}$	$2.1 \cdot 10^{-7}$	$5.9 \cdot 10^{-7}$	$4.4 \cdot 10^{-6}$	$3.8 \cdot 10^{-6}$
Avg. EOC	**4.2**	**3.75**	**4.1**	**2.8**	**3.3**

Application: Euler-Gravity

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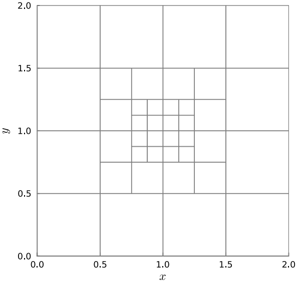

N	$L^1(\rho)$	$L^1(v_{x,y})$	$L^1(p)$	$L^2(\phi)$	$L^2(q_{1,2})$
40	$2.8 \cdot 10^{-4}$	$7.8 \cdot 10^{-5}$	$3.5 \cdot 10^{-4}$	$4.4 \cdot 10^{-4}$	$7.1 \cdot 10^{-4}$
160	$1.5 \cdot 10^{-5}$	$5.6 \cdot 10^{-6}$	$1.9 \cdot 10^{-5}$	$1.4 \cdot 10^{-5}$	$5.0 \cdot 10^{-5}$
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Application: Euler-Gravity

Jeans Gravitational Instability[1]

- Stable oscillation of a self-gravitating gas cloud
- Analytical expressions for pot., kin., int. energies

[1] Binney, Tremaine; Galactic Dynamics; 2011.

Application: Euler-Gravity

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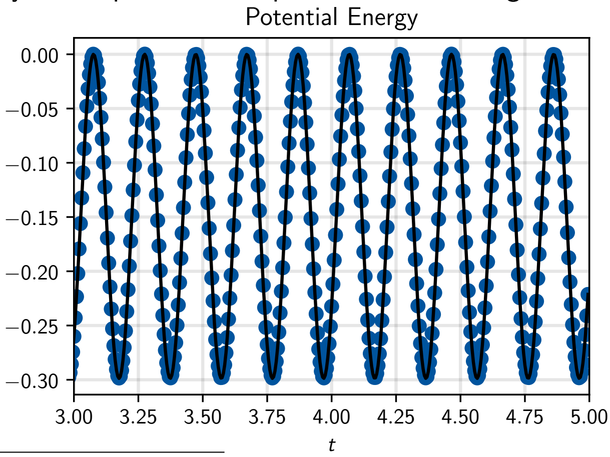

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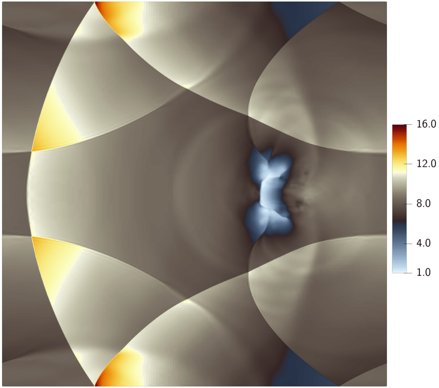
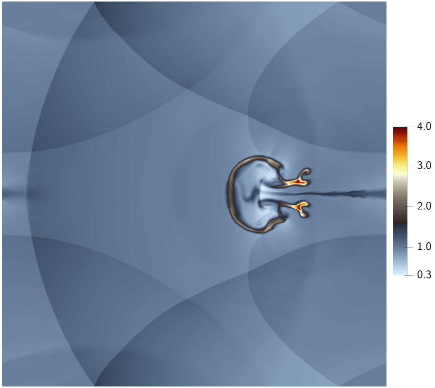
Application: Euler-Gravity

Blob Test[1] + Self-Gravity



Pressure p



Density ρ

[1] Aretz et al.; Mon. Not. R. Astron. Soc; 2006.

Application: Euler-Gravity

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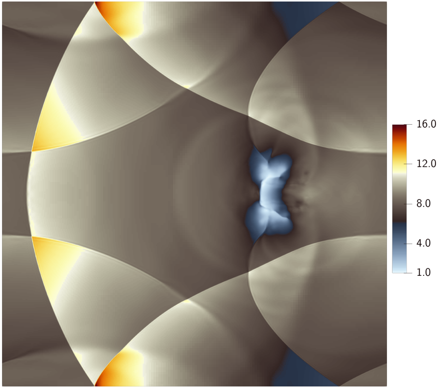
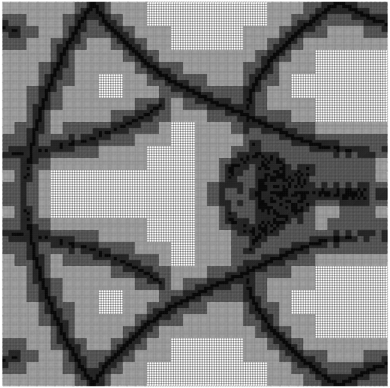


Pressure p



Mesh

[1] Aretz et. al; Mon. Not. R. Astron. Soc; 2006.